Heat Transfer in a Stirred Tank

Group # 1

Diego Martinez

Allison Nordvall

Peter Starakiewicz

Emily Sobie

James Fell

William Rawson

Chemical Engineering Lab II

2/4/2020

UIC UNIVERSITY OF ILLINOIS AT CHICAGO

Introduction

- **Objective**
  - The objective of this experiment is to study a selection of steady state heat transfer processes. The emphasis of this study will be to measure the heat transfer coefficient between the water and the inside vessel wall of the continuously stirred tank.
  - Parameters of the study will be manipulated to observe the dependence of the heat transfer coefficient on fluid properties, impeller speeds, and in the presence of baffles.

Theory

- **Heat transfer can help by convection or conduction.**
  - **Convection heat transfer is when energy is transfer by bulk flow of the fluid.**
    - $Q = Ah(T_s - T_f)$ **(1)**
  - **Conductive heat transfer is when energy is transferred by colliding particles and movement of electrons within a body.**
    - $Q = kA\frac{dT}{dx}$
- **At steady-steady state heat flux does not change over time.**
  - $Q = kA\frac{(T_{s1} - T_{s2})}{x_{length}} = Ah\left(T_{s2} - T_f\right) = UA(T_{s1} - T_f)$

UIC UNIVERSITY OF ILLINOIS AT CHICAGO

Theory

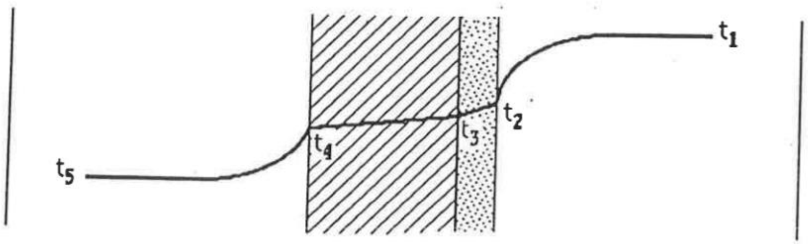

Figure 1. Conductive and Convective heat transfer

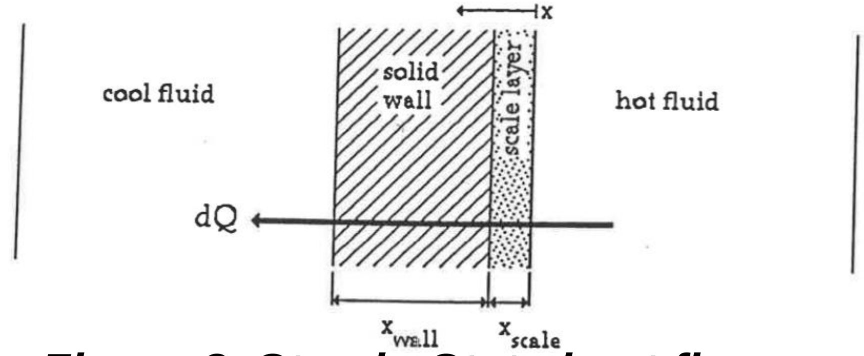

Figure 2. Steady-State heat flux

UIC UNIVERSITY OF ILLINOIS AT CHICAGO

Apparatus

- **This lab focused on the continuously stirred tank (CST), key components include:**
  - **Internal Cooling Coil**
  - **Baffles**
  - **Pump**
  - **Heat Exchanger**
  - **Flow Controllers**
  - **Steam Jacket**
  - **Impeller**
  - **RPM Controller/Display**
  - **Drain**
  - **Steam Condenser**
  - **Valves for steam, cooling water and tank water inlet**
  - **Temperature Gauges**
  - **Steam Pressure Gauge**

Material & Supplies

- Water
- Steam
- Graduated Cylinder
- Stopwatch
- Safety Goggles
- Heat Resistant Gloves

Procedure

Choose two of the three experiments to do during the lab.
A: Steady state for an unbaffled tank,
B: Steady state for a baffled tank
C: unsteady state for an unbaffled tank.

General Operating Procedure:
1.) Fill the tank 2 inches from the top and turn on impeller to 650 RPM.
2.) Turn on the coiling coil (open valve fully) and turn on cooling water for heat exchanger.
3.) Make sure discharge valve is closed fully, and suction and bypass valves are open.
4.) Turn pump on, run for 5 seconds, slowly open discharge valve and close bypass.
5.) Open steam valve
6.) Follow procedure for A, B or C. For these parts of the experiments you will be changing flowrates of the coiling coil and heat exchanger cooling water or will be changing the impeller speed. These are your manipulated variables that are being studied for correlations between them and the heat transfer coefficient.

Experimental Challenges

- Heating the tank to 65 ^{o}C, requires ~ ¼ of a turn for the steam valve and ~ ½ of a turn for > 70 ^{o}C.
- Open the cooling coil flow valve all the way. Slowly open the cooling water valve to the heat exchanger until the flow through the cooling coil starts to decrease and then stop opening the valve.
- Overshoot the RPM of the impeller by 25 RPM of the desired RPM. The RPM of the impeller drops after about 30 seconds by 25 and fluctuates by +/- 10 RPM of the desired set point.
- Cooling coil affects the temperature of the tank much greater than the heat exchanger.

Safety

- **Potential Electrical Hazards**
  - There are electrical hazards in this experiment which are the mixer, pump, and the RPM reader. Mixing this electronic equipment with the water in this experiment could lead to shock.
- **Potential Mechanical hazards**
  - A spinning impeller and a running motor can cause pinch points. This can potentially cause bodily harm.
- **Potential Pressure Hazards**
  - If temperature of the vessel is not controlled and the steam piping could have a pressure increase.
  - Max expected at 11psig
  - If pressure of steam increases, there is a safety release valve on the stream.
- **Potential Temperature Hazards**
  - Hot spots in the system like the vessel and hot streams.
  - Heat resistant gloves need.
  - Vessel is expected to reach 125 ℃
  - Pump switch, impeller control switch, chill water valves, drained valve, and steam valve can used to reduce heat.
- **Hazardous/Toxic Chemicals**
  - None